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SOLUTIONS

Holy Cross College

Semester 2 Examination 2012

Question/Answer Booklet

Year 10 Science (Modified)

Please place your student identification label in this box

Student Name ____________________

Student's Teacher ____________________

Time allowed for this paper

Reading time before commencing work: 10 minutes
Working time for paper: 100 minutes

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Data Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid, ruler, highlighters.

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course.

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple Choice	55	55	40	55	38
Section Two: Short Answer	16	16	45	75	51
Section Three: Comprehension	9	9	15	17	11
				147	100

Instructions to candidates

1. The rules for the conduct of examinations at Holy Cross College are detailed in the College Examination Policy. Sitting this examination implies that you agree to abide by these rules.

2. Write your answers in this Question/Answer Booklet.

3. Working or reasoning should be clearly shown when calculating or estimating answers.

4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Achievement Standards

The atomic structure and properties of elements are used to organise them in the Periodic Table.

Different types of chemical reactions are used to produce a range of products and can occur at different rates.

The motion of objects can be described and predicted using the laws of physics.

SECTION 1 MULTIPLE CHOICE (55 marks)

Choose the best answer for each question in the Multiple Choice Question booklet.

Mark your selection with a cross.

1	(a)	(b)	(c)	(d)
2	(a)	(b)	(c)	(d)
3	(a)	(b)	(c)	(d)
4	(a)	(b)	(c)	(d)
5	(a)	(b)	(c)	(d)
6	(a)	(b)	(c)	(d)
7	(a)	(b)	(c)	(d)
8	(a)	(b)	(c)	(d)
9	(a)	(b)	(c)	(d)
10	(a)	(b)	(c)	(d)
11	(a)	(b)	(c)	(d)
12	(a)	(b)	(c)	(d)
13	(a)	(b)	(c)	(d)
14	(a)	(b)	(c)	(d)
15	(a)	(b)	(c)	(d)
16	(a)	(b)	(c)	(d)
17	(a)	(b)	(c)	(d)
18	(a)	(b)	(c)	(d)
19	(a)	(b)	(c)	(d)
20	(a)	(b)	(c)	(d)
21	(a)	(b)	(c)	(d)
22	(a)	(b)	(c)	(d)
23	(a)	(b)	(c)	(d)
24	(a)	(b)	(c)	(d)
25	(a)	(b)	(c)	(d)
26	(a)	(b)	(c)	(d)
27	(a)	(b)	(c)	(d)
28	(a)	(b)	(c)	(d)
29	(a)	(b)	(c)	(d)
30	(a)	(b)	(c)	(d)
31	(a)	(b)	(c)	(d)
32	(a)	(b)	(c)	(d)
33	(a)	(b)	(c)	(d)
34	(a)	(b)	(c)	(d)
35	(a)	(b)	(c)	(d)
~~36~~	~~(a)~~	~~(b)~~	~~(c)~~	~~(d)~~
37	(a)	(b)	(c)	(d)
38	(a)	(b)	(c)	(d)
39	(a)	(b)	(c)	(d)
40	(a)	(b)	(c)	(d)
41	(a)	(b)	(c)	(d)
42	(a)	(b)	(c)	(d)
43	(a)	(b)	(c)	(d)
44	(a)	(b)	(c)	(d)
45	(a)	(b)	(c)	(d)
46	(a)	(b)	(c)	(d)
47	(a)	(b)	(c)	(d)
48	(a)	(b)	(c)	(d)
49	(a)	(b)	(c)	(d)
50	(a)	(b)	(c)	(d)
51	(a)	(b)	(c)	(d)
52	(a)	(b)	(c)	(d)
53	(a)	(b)	(c)	(d)
54	(a)	(b)	(c)	(d)
55	(a)	(b)	(c)	(d)

SECTION 2 SHORT ANSWER (75 marks)

Attempt all questions. Write your answers in the spaces provided.

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1. Classify the following elements as ***alkali metals, transition metals, metalloid (semi-metals), halogens*** or ***noble (inert) gases***. Use the periodic table on the data sheet provided.

K Ge He Cu Fr Cl Ag Ar

Alkali Metal	**Transition Metal**	**Metalloid (Semi-Metal)**	**Halogen**	**Noble (Inert) Gas**
K Fr	Cu Ag	Ge	Cl	He Ar

(4 marks)

2. Complete the following table showing the constituent particles of each element.

ELEMENT	**PROTONS**	**NEUTRONS**	**ELECTRONS**
$^{28}_{14}Si$	14	14	14
$^{39}_{19}K$	19	20	19
$^{14}_{7}N$	7	7	7

($\frac{1}{2}$ mark off each mistake.)

(4 marks)

3. Draw clear diagrams to show the electron configurations of the following elements. ***Remember to include the number of particles in the nucleus.***

(a) $^{12}_{6}C$

6p 6n 2 4

(b) $^{40}_{18}Ar$

18p 20n 2 8 8

(c) $^{27}_{13}Al$

13p 14n 2 8 3

[Nucleus - 1 mark
Electrons - 1 mark]

(6 marks)

4. Write the name for each of the following compounds.
(Note Some of them are ionic and you need your list of ions.)

FORMULA	NAME
KNO_3	potassium nitrate
$Mg_3(PO_4)_2$	magnesium phosphate
$Al_2(SO_4)_3$	aluminium sulphate
PBr_3	phosphorus tribromide
FeI_3	iron iodide
$Ba(CH_3COO)_2$	barium acetate
S_2I_4	disulphur tetraiodide

(1 mark each).

(7 marks)

5. Balance the following equations.

(a) $2\ Al + 6\ HCl \rightarrow 2\ AlCl_3 + 3\ H_2$ (2 marks each)

(b) $C_3H_8 + 5\ O_2 \rightarrow 3\ CO_2 + 4\ H_2O$

(c) $2\ NaOH + Fe(NO_3)_2 \rightarrow Fe(OH)_2 + 2\ NaNO_3$

(6 marks)

6. In the table below, choose the correct meaning for each word given. ***Write the letter of the correct meaning next to the word.***

WORD	CORRECT LETTER	MEANINGS
neutron	F	A - a row of the periodic table.
period	A	B - building block of matter.
molecule	D	C - two or more elements chemically combined.
compound	C	D - two or more atoms chemically combined.
group	E	E - column of the periodic table.
atom	B	F - particle with no charge.

($\frac{1}{2}$ mark each)

(3 marks)

7. Write complete word equations for the following acid reactions.

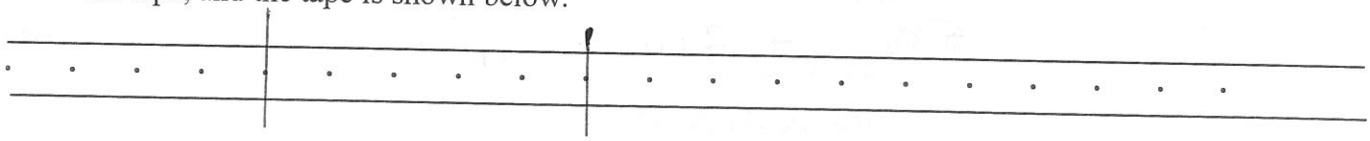
(a) zinc carbonate + nitric acid → zinc nitrate + water + carbon dioxide. (2)

(b) potassium hydroxide + sulphuric acid → potassium sulphate + water (2)

(c) calcium carbonate + hydrochloric acid → calcium chloride + water + carbon dioxide. (2)

(6 marks)

8. Some students investigated the movement of a dynamics trolley across a bench using a ticker timer to record the motion. The timer puts 50 dots per second (1 dot-spacing = 0.02 sec) onto the tape, and the tape is shown below.



(a) Using any 6 dots of your choosing, perform a calculation of the ***average speed***. Show all of your working below.

(1)

d = 4·2 cm = 0·042 m (Remember to convert.) (1 mark)

t = 0·1 s (1) (1 mark)

(Calculation: use speed, d, t.)

$$\text{speed} = \frac{d}{t}$$

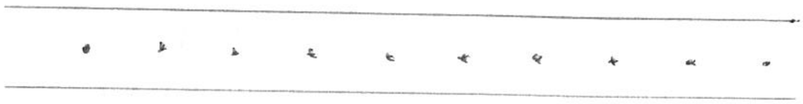
$$= \frac{0{\cdot}042}{0{\cdot}1} \quad (1)$$

$$= 0{\cdot}42\ ms^{-1} \quad (1)$$

(2 marks)

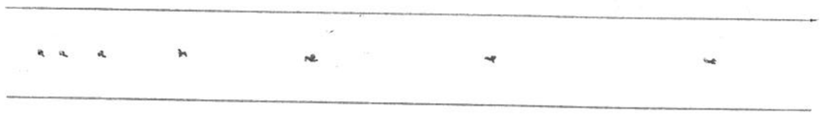
(b) What is the difference between a ticker tape for constant velocity and acceleration? ***Draw a representation of each tape below.***

(i) Constant Velocity



(1)

(ii) Acceleration



(1)

(2 marks)

9. Vincent dropped a stone down a well and saw it hit the bottom 3.5 s later. Calculate the velocity of the stone at impact, given the acceleration due to gravity is 9.80 ms^{-2} and the stone starts from rest when it is dropped.
(Hint: Use v, u, a, t.)

$\downarrow$ +ve

$v = ?$

$u = 0\ ms^{-1}$

$a = 9{\cdot}80\ ms^{-2}$

$t = 3{\cdot}5\ s$

$v = u + at$ (1)

$= 0 + (9{\cdot}80)(3{\cdot}50)$ (1)

$= \underline{34{\cdot}3\ ms^{-1}\text{ down}}$ (1)

(3 marks)

10. Sean steps on the brake pedal of his car and applies a regarding force of 2100 N to slow his 750 kg car. Calculate his car's deceleration.
(Hint: Use F, m, a.)

Forwards +ve

$F = -2100N$

$m = 750\ kg$

$a = ?$

$a = \frac{F}{m}$ (1)

$= \frac{-2100}{750}$ (1)

$= \underline{-2{\cdot}8\ ms^{-2}\text{ backwards}}$ (1)

(3 marks)

11. Max has a mass of 57 kg on Earth. He is lucky enough to be selected for a junior astronaut programme and journeys to the International Space Station (ISS) for a 3-month stay.

(a) Calculate Max's weight on Earth.
(Hint: Use F_w, m, g.)

$F_w = ?$

$m = 57\ kg$

$g = 9.80\ ms^{-2}$

$F_w = mg$ (1)

$= (57)(9.80)$ (1)

$= \underline{558.6\ N}$ (1)

(3 marks)

(b) Max "floats" in space within the ISS. Is he weightless? Explain your answer.

- no (1)
- Gravity of the Earth is pulling on him. (1)

(2 marks)

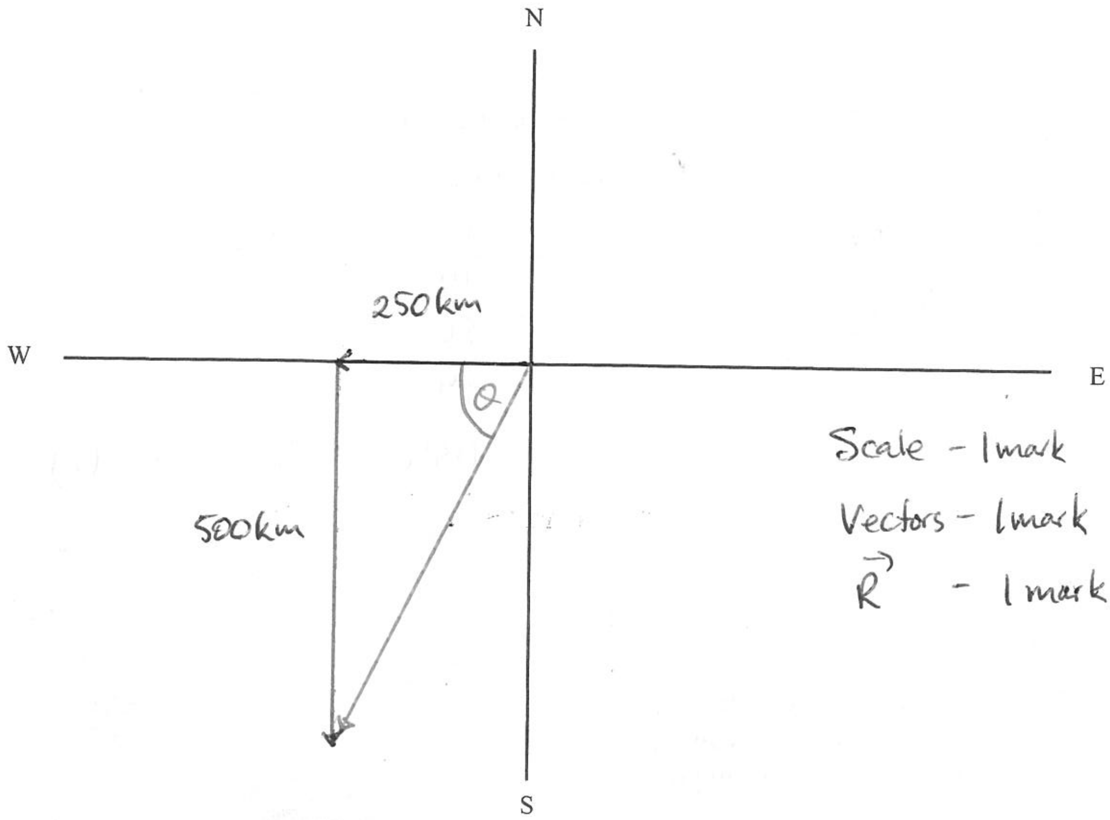
(c) What is Max's mass in the space station?

57 kg (1)

(1 mark)

12. A plane takes off from a country airstrip and flies 250 km West, then 500 km South. It lands on a farm owned by the pilot. The total time of flight is 1.5 hours.

(a) Draw a vector diagram to show the addition of the vectors and the resultant vector. ***(Use a scale of 1.0 cm = 100 km.)***



Scale - 1 mark

Vectors - 1 mark

$\vec{R}$ - 1 mark

(3 marks)

(b) Calculate the ***displacement*** of the plane from its starting point by ***measuring the length*** of the resultant vector. Use a protractor to find the ***direction*** of the resultant.

$\vec{R}$ = 570 km (1)

θ = 60° (1)

$\vec{R}$ = 570 km W 60° S. (1)

(3 marks)

(c) Calculate the ***distance*** travelled by the plane for the journey. ***Give your answer in km.***

$d = 250 + 500$ (1)

$= \underline{750 \text{ km}}$ (1)

(2 marks)

13. A force of 36 N acts on a mass of 24 kg. What is the acceleration of the mass? (Hint: Use F, m, a.)

Forwards +ve

$F = 36\text{N}$

$m = 24\text{kg}$

$a = ?$

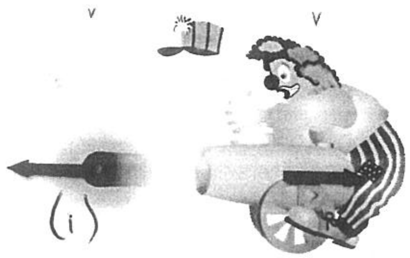
$a = \frac{F}{m}$ (1)

$= \frac{36}{24}$ (1)

$= \underline{1.5 \text{ ms}^{-2} \text{ forwards}}$ (1)

(3 marks)

14. Look at the diagram showing a clown and a cannon at a circus. Analyse it and explain why the cannon hits the clown, using Newton's Laws in your explanation.



- Action – cannon ball moves forwards. (1)
- Reaction – cannon moves backwards. (1)

(2 marks)

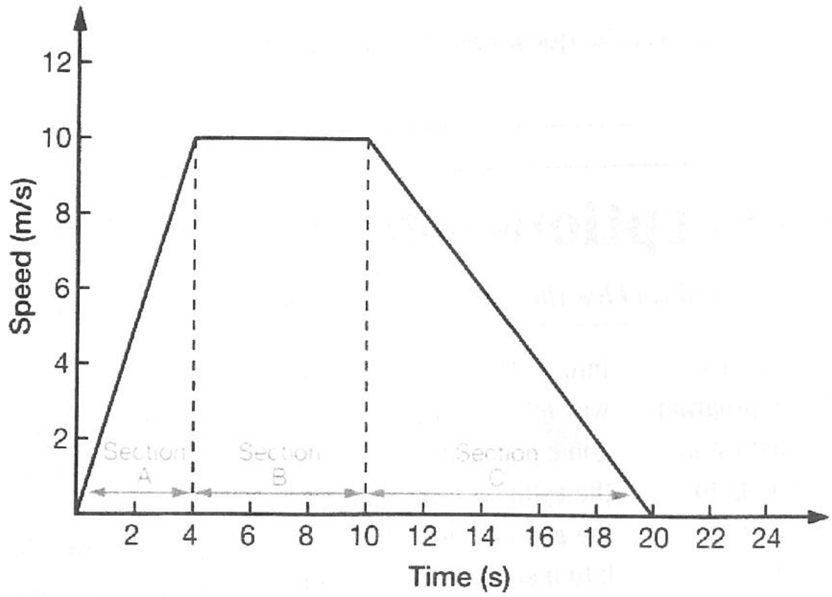
15. What is the resultant force acting on the parachutist?

$\Sigma F = 800 - 800$ (1)

$= \underline{0 \text{ N}}.$ (1)

(2 marks)

16. The graph below shows the motion of a bicycle over a 20.0 s period.



(a) During which time intervals was the bicycle travelling at a constant velocity?

4-10 s . (2).

(2 marks)

(b) From the area under the graph, determine the ***displacement*** of the bicycle for the 20.0 s. ***(Show all of your working.)***

s = area

$= \frac{1}{2}(4)(10) + (6)(10) + \frac{1}{2}(10)(10)$ (3)

$= \underline{130 \text{ m}}$ (1)

(4 marks)

SECTION 3 COMPREHENSION (17 marks)

Read the article and then answer the questions that follow. Write your answers in the spaces provided.

Mythconceptions—fry me to the moon

Dr Karl S. Kruszelnicki tackles the myths, curiosities and absurdities of everyday life.

One of the great myths about the space program that got us to the moon was that 'the only good thing to come out of it was Teflon'.

Teflon was in fact discovered, albeit accidentally, much earlier, on April 6, 1938, when a Du Pont chemist, Roy J Plunkett Jr, was looking for a new gas to use in refrigerators. One gas on his list was tetrafluoroethylene. Plunkett opened the valve of what was supposed to be a full cylinder of it, but none came out.

Most people would have just fetched another cylinder, but Plunkett was curious.

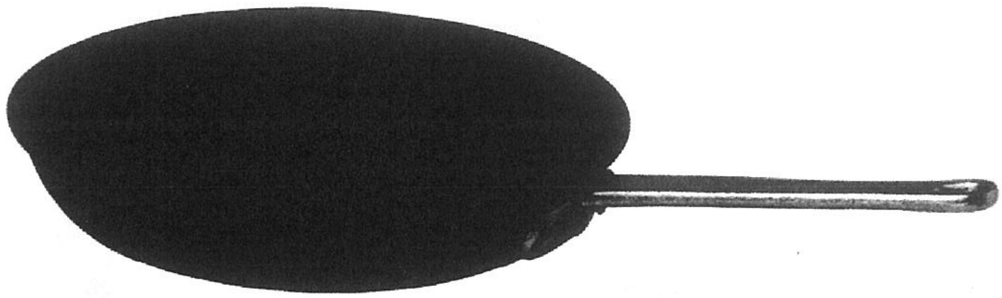
He weighed the cylinder and found that it weighed as much as a full cylinder of tetrafluoroethylene, so it wasn't empty. He then proved that the valve was indeed open by running a thin wire into it. There was only one way left to work out what was going on, so he hacksawed the cylinder open, and found the gas had turned itself into a solid, white, slippery, waxy powder which, when analysed, revealed itself as polytetrafluoroethylene, today known as Teflon.

This powder proved to be the most inert plastic Plunkett had ever come across. It was unaffected by low heat. Nothing dissolved it. And it was more slippery than wet ice on wet ice. It was also a terrific electrical insulator. Sadly, it was very expensive to produce, so it had no commercial use.

During World War II, the Manhattan Project saw the first atom bombs made. As part of the manufacturing process, uranium had to be turned into uranium hexafluoride. Unfortunately, this was an incredibly corrosive gas and would eat through metal, a major problem. But the Manhattan Project had money to burn, and once the US military heard about Teflon, it bought huge quantities of it to protect the machinery from the corrosive uranium hexafluoride.

But Teflon would also have applications in other important areas.

The human immune system tends to ignore Teflon, so it's used in artificial heart valves and aortas, as well as for artificial corneas and bones (for ears, chins, fingers, noses, skulls and various joints). Because it's tough enough to withstand the harsh conditions of outer space, Teflon is also used in spacesuits. And, of course, it's there in frying pans.



1. When was Teflon discovered and by whom?

April 6 1938 (1) – Roy Plunkett Jnr. (1)

(2 marks)

2. What is the correct chemical name of Teflon?

polytetrafluoroethylene. (1)

(1 mark)

3. Describe what Teflon looks like.

Solid, white, slippery, waxy powder.
(1) (1)

(2 marks)

4. What properties make Teflon an extremely useful material?

- Unaffected by low heat.
- Nothing dissolves it.
- More slippery than wet ice.
- Electrical insulator.

(Any 2 - 2marks).

(2 marks)

5. Teflon is said to be inert - what does this mean?

Won't react. (1).

(1 mark)

6. What research was the inventor doing when he discovered Teflon?

Looking for a new gas to use in refrigerators.

(1 mark)

7. What stopped Teflon from being commercially produced in the early days?

Very expensive to produce.

(1 mark)

8. Is Teflon a metal or a plastic? Give one piece of evidence.

plastic. (½)
Electrical insulator. (½)

(1 mark)

9. Complete the following table about the uses of Teflon.

LOCATION	USE	PROPERTY RELATED TO USES
Body	Heart valves aorta corneas bones	Immune system ignores it (unreactive)
Space	space suit	Tough enough to withstand harsh conditions.
Kitchen	frying pans.	Slippery unreactive.

(6 marks)

END OF EXAMINATION

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